Tuan N. Le

Master's Student – Astronomy & Astrophysics
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Education

University of Science, VNUHCM (HCMUS) 2025 – present
M.Sc. in Theoretical Physics *Ho Chi Minh City, Vietnam*

University of Technology, VNUHCM (HCMUT) 2019 – 2023
B.Eng. in Mechanical Engineering *Ho Chi Minh City, Vietnam*

Research Interests & Skills

- **Topics:** galaxy dynamics, supermassive and intermediate-mass black hole mass measurements, stellar and gas dynamical modeling, black hole–host galaxy coevolution
- **Methods:** Jeans Anisotropic Modeling (JAM), Multi-Gaussian Expansion (MGE), Penalized Pixel-Fitting (pPXF), MCMC / Bayesian inference, integral-field spectroscopy
- **Programming:** Python (NumPy, SciPy, Matplotlib, Astropy); CapFit

Research Experience

Measuring the IMBH in IC 342 using Keck/OSIRIS 2026 – present
Project lead · First author · Advisor: Dr. Dieu D. Nguyen *Paper in prep.*

- Lead the measurement of the central intermediate-mass black hole in IC 342, a nearby spiral seen through heavy dust extinction near the Galactic plane, to probe the black hole–host galaxy relation at the low-mass end
- Build the stellar mass model via Multi-Gaussian Expansion (MGE) of HST imaging and extract line-of-sight kinematics from Keck/OSIRIS CO-bandhead spectra with pPXF
- Apply Jeans Anisotropic Modeling (JAM) with MCMC sampling to derive posterior constraints on the black hole mass, mass-to-light ratio, and orbital anisotropy

Black hole masses from cold-gas kinematics at radio wavelengths 2025-2026
Project lead · First author · Advisor: Dr. Dieu D. Nguyen *Submitted, AAS Journals*

- Determined the central black hole mass in NGC 3862 from cold molecular-gas kinematics, a more reliable tracer than hot ionized gas for probing galaxy–black hole coevolution
- Modeled the gas rotation field to constrain the dynamical black hole mass
- Separated the non-thermal synchrotron emission from the bright central jet out of the photometry, removing its contribution to the nuclear point source so that the stellar mass model (MGE) is not biased

Weighing the central SMBH in M81 (NGC 3031) with JWST/NIRSpec 2025-2026
Lead contributor · Co-author · Advisor: Dr. Dieu D. Nguyen *Published, ApJ Journals*

- Delivered the first robust stellar-dynamical mass of M81's SMBH from high-resolution 2D JWST/NIRSpec kinematics, resolving long-standing doubts over previous measurements; near-infrared data penetrate the nuclear dust and separate stellar light from the non-thermal AGN continuum
- Modeled the kinematics with Jeans Anisotropic Modeling (JAM), constraining $M_{\rm BH} = (4.77 \pm 0.37) \times 10^7\, M_\odot$ (1σ, incl. systematics)
- Quantified systematic uncertainties via an ensemble of 24 independent models spanning different M/L gradients, PSF, AGN masking, and velocity-ellipsoid assumptions, then bootstrapped the resulting $M_{\rm BH}$ values

APPOINTMENTS

Research Assistant, ICISE 2025
Quy Nhon, Vietnam *Advisor: Dr. Dieu D. Nguyen*

- Contributed to multiple supermassive/intermediate-mass black hole mass-measurement projects using stellar and gas dynamical modeling (see Research Experience)
- Co-developed observing/telescope-time proposals and supporting analyses for JWST
- Built Python pipelines for reduction and analysis of integral-field kinematic data from JWST/NIRSpec

PUBLICATIONS

($\star$ = first author. Full author lists; my name in **bold**.)

Peer-Reviewed

1. D. D. Nguyen, **T. N. Le**, M. Cappellari, H. N. Ngo, T. Q. T. Le, T. H. T. Ho, L. Q. T. Nguyen, E. Gallo, F. Zou, M. Perna, N. Thatte, M. Pereira-Santaella, *The Supermassive Black Hole in the Nearby Spiral Galaxy M81: A Robust Mass from JWST/NIRSpec Stellar Dynamics*, ApJ 1003, 98 (2026). doi:10.3847/1538-4357/ae64f0

2. D. D. Nguyen, H. N. Ngo, M. Cappellari, T. Q. T. Le, T. H. T. Ho, **T. N. Le**, E. Gallo, F. Zou, N. Thatte, M. Perna, M. Pereira-Santaella, *Measuring the Central Dark Mass in NGC 4258 with JWST/NIRSpec Stellar Kinematics*, ApJ 999, 97 (2026). doi:10.3847/1538-4357/ae32f2

3. D. D. Nguyen, M. Cappellari, H. N. Ngo, T. Q. T. Le, **T. N. Le**, K. N. H. Ho, A. K. Nguyen, P. T. On, H. G. Tong, N. Thatte, *Simulating Intermediate Black Hole Mass Measurements for a Sample of Galaxies with Nuclear Star Clusters using ELT/HARMONI High Spatial Resolution Integral-Field Stellar Kinematics*, AJ 170, 124 (2025). doi:10.3847/1538-3881/ade9ba

Submitted

1. $\star$**T. N. Le**, D. D. Nguyen, A. W. Graham, M. Durré, T. H. T. Ho, H. N. Ngo, E. Gallo, I. V. Chilingarian, T. Q. T. Le, F. Zou, R. Soria, *A $10^5\,M_\odot$ Black Hole in the Bulgeless Spiral Galaxy IC 342: First Stellar-Dynamical $M_{\rm BH}$ Measurement with Keck/*OSIRIS *LGS-AO*, submitted to AAS Journals.

2. $\star$**T. N. Le**, D. D. Nguyen, H. N. Ngo, E. Gallo, T. Q. T. Le, T. H. T. Ho, F. Zou, Q. T. Le, *Unveiling the Central Black Hole of NGC 3862: A Direct Mass Determination Using ALMA's High-Resolution Molecular Gas Kinematics*, submitted to AAS Journals.

In Preparation

1. $\star$**T. N. Le**

PRESENTATIONS

- **Speaker** – *Uncovering Intermediate-Mass Black Holes in Extragalactic Globular Clusters using ELT/HARMONI High Spatial Resolution Integral-Field Stellar Kinematics*. SAGI Astrophysics Workshop on Dust Polarimetry, ICISE, Vietnam (2023).

TRAINING & SCHOOLS

- **Participant** – 48th International School for Young Astronomers (ISYA), ICISE, Vietnam (2026).
- **Participant** – 10th Vietnam School of Astrophysics (VSOA10), track: Cosmology. ICISE, Vietnam (2026).
- **Participant** – School on Quantum Optimization Algorithms (SQOA), Phan Thiet, Vietnam (2026).
- **Participant** – 9th Vietnam School of Astrophysics (VSOA9), track: Galaxies. ICISE, Vietnam (2025).

TEACHING & OUTREACH

- **Mathematics Tutor** – Taught mathematics to high-school students (2025–present).
- **Event Host & Moderator** – *PhD Application Journey in Astronomy for Students*, [University of Science Astronomy Club] (2026).

References

Dr. Dieu D. Nguyen
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